CSA0412-OPERATING SYSTEMS

EXP 06-Preemptive priority scheduling algorithm

```
public class Main {
    public static void main(String[] args) {
        int n = 4;

        int[] at = {0, 1, 2, 3};       // Arrival time
        int[] bt = {8, 4, 2, 1};       // Burst time
        int[] pr = {2, 1, 3, 2};       // Priority

        int[] rt = bt.clone();         // Remaining time
        int[] ct = new int[n];         // Completion time

        int time = 0, completed = 0;

        while (completed < n) {
            int idx = -1;
            int best = Integer.MAX_VALUE;

            for (int i = 0; i < n; i++) {
                if (at[i] <= time && rt[i] > 0
                        && pr[i] < best) {
                    best = pr[i];
                    idx = i;
                }
            }

            if (idx == -1) {
                time++;
```

```
            continue;
        }

        rt[idx]--;
        time++;

        if (rt[idx] == 0) {
            ct[idx] = time;
            completed++;
        }
    }

    System.out.println("PID\tAT\tBT\tPriority\tCT\tTAT\tWT");

    for (int i = 0; i < n; i++) {
        int tat = ct[i] - at[i];
        int wt = tat - bt[i];

        System.out.println("P" + (i + 1) + "\t"
                + at[i] + "\t" + bt[i] + "\t"
                + pr[i] + "\t\t" + ct[i] + "\t"
                + tat + "\t" + wt);
    }
  }
}
```

Output

```
PID AT  BT  Priority    CT  TAT WT
P1  0   8   2       12  12  4
P2  1   4   1       5   4   0
P3  2   2   3       15  13  11
P4  3   1   2       13  10  9
```